

Employee Attrition Analysis using Machine Learning

Internship Project • Week 2 Report | Kunal Nigewan

Objective

Analyzed employee attrition patterns to identify the key factors that influence employees leaving an organization. The project focused on understanding workplace behavior, satisfaction trends, and retention drivers using data analysis and machine learning.

Procedure

- Loaded and explored the HR Attrition dataset using Pandas.
- Checked for missing values and verified data quality.
- Removed unnecessary columns (EmployeeNumber, Over18, StandardHours).
- Converted categorical columns to numeric using One-Hot Encoding.
- Scaled numerical features using StandardScaler.
- Performed exploratory data analysis with 10+ visualizations.
- Trained Logistic Regression, Random Forest, and Gradient Boosting models.
- Compared models using accuracy, recall, precision, and F1-score.

Model Results

Model	Accuracy	Key Observation
Logistic Regression	71.7%	Better at identifying employees likely to leave
Random Forest	88.0%	Highest overall prediction accuracy
Gradient Boosting	87.0%	Balanced performance across all metrics

Charts & Insights

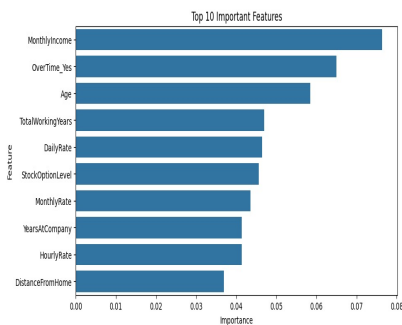


Fig 1: Top 10 Important Features — Monthly Income, Overtime, and Age are the strongest attrition predictors.

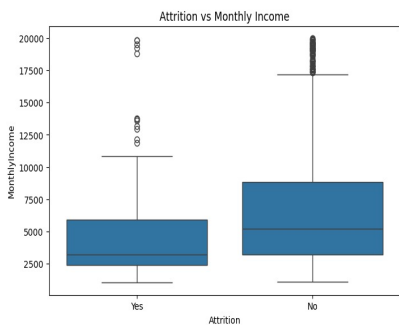


Fig 2: Attrition vs Monthly Income — employees who left had noticeably lower median income.

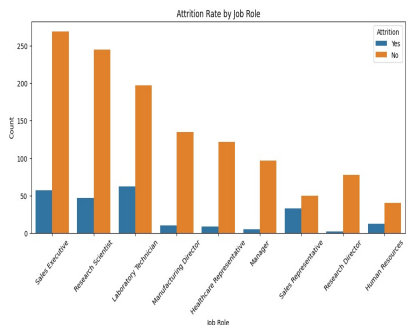


Fig 3: Attrition by Job Role — Sales Executives and Research Scientists show the highest absolute attrition counts.

Key Findings

- Employees working overtime showed significantly higher attrition rates.
- Lower monthly income was a strong indicator of employees likely to leave.
- Sales Representatives and Laboratory Technicians had higher turnover than other roles.
- Poor work-life balance and lower job satisfaction correlated with higher exit rates.
- Most resignations occurred during the early years of employment.
- Younger employees showed slightly higher attrition compared to older employees.

HR Recommendations

- Monitor employee workload and overtime to reduce burnout.

- Introduce engagement programs for newly joined employees to improve early-stage retention.
- Conduct regular feedback and satisfaction surveys to identify workplace concerns early.
- Offer career development and skill growth initiatives to improve long-term retention.
- Review high-attribution job roles more closely to understand employee concerns.
- Promote better work-life balance practices across all departments.

Conclusion

This project successfully identified the key factors driving employee attrition and provided actionable HR insights. The findings demonstrate how data-driven analysis can help organizations understand employee behavior, improve workplace satisfaction, and build stronger long-term retention strategies.